Haloalkanes and Haloarenes

CLASS OBJECTIVES

Recap

Preparation of haloalkanes and haloarenes

Physical Properties of Haloalkanes and Haloarenes

REVISION

Which of the following is a secondary alkyl halide ?

(a) Isobutyl chloride
(b) Isopentyl chloride
(c) Neopentyl chloride
(d) Isopropyl chloride

(d) Isopropyl chloride

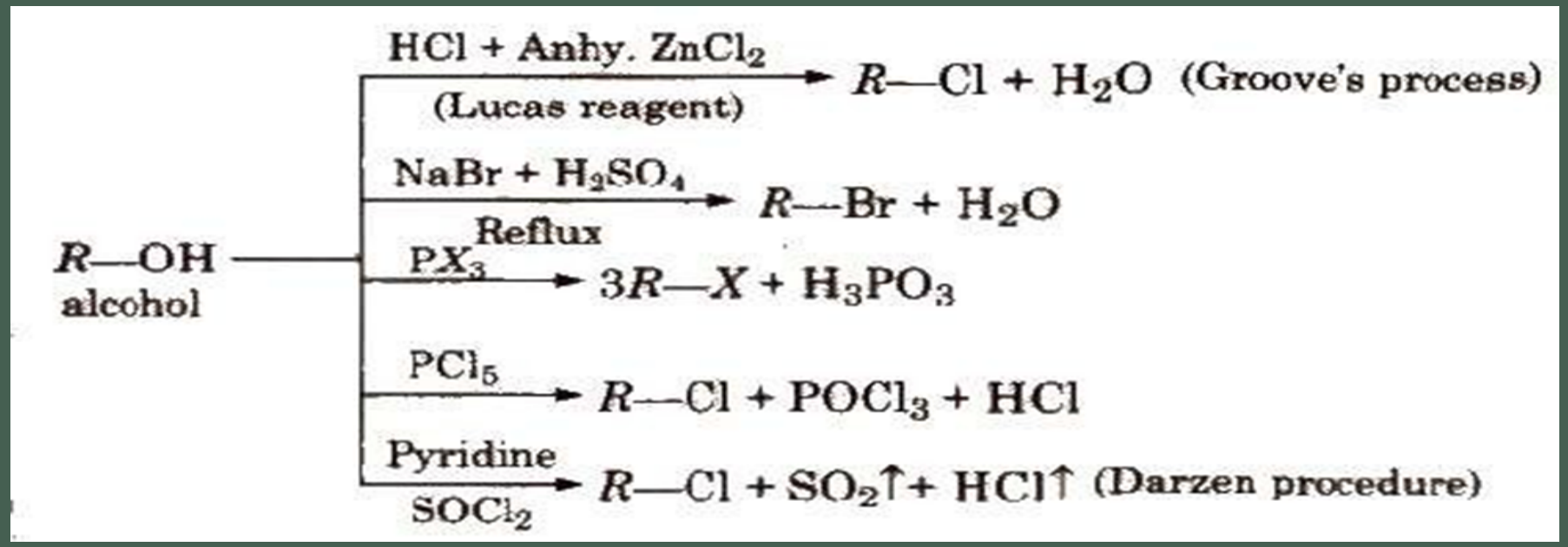
Revision

The IUPAC name of the compound CH3-CH=CH-CH2-Br is

(a) 4-Bromobut-2-ene
(b) 1-Bromobut-2-ene
(c) 3-Bromobut-2-ene
(d) Allyl bromide

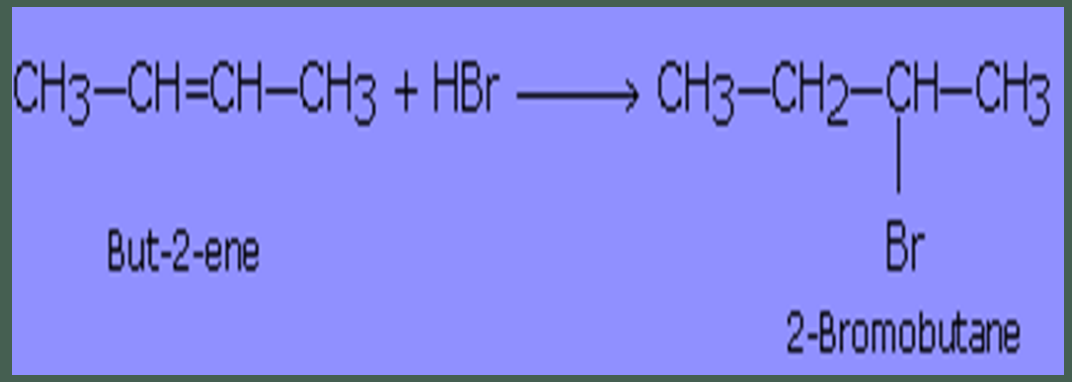
Ans (b) 1-Bromobut-2-ene

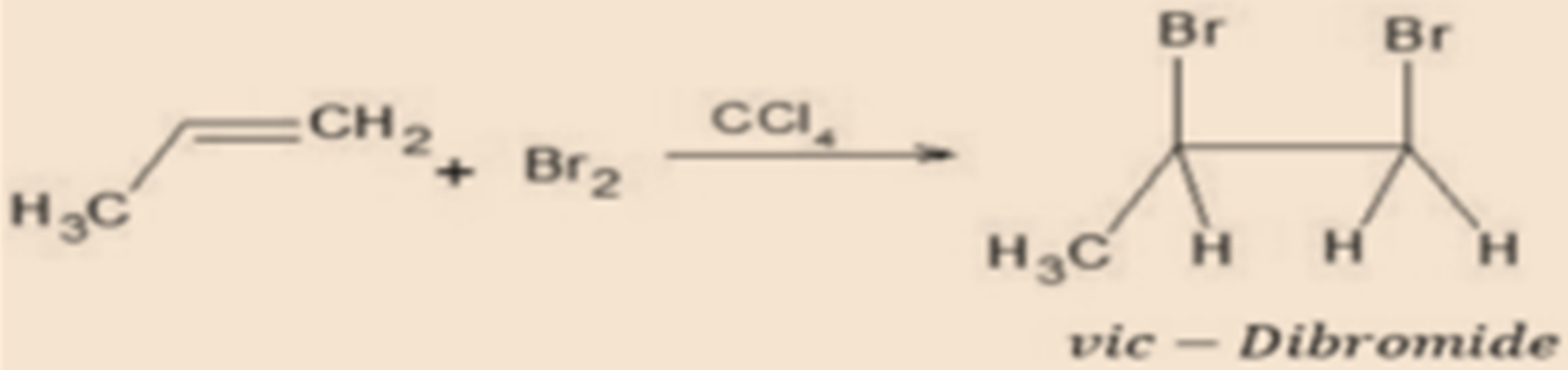
Preparations of Haloalkanes



$$CH_3CH_2CH_2CH_3 \xrightarrow{Cl_2/h\nu} CH_3CH_2CH_2CH_2Cl + CH_3CH_2CH(Cl)CH_3$$

Free Radical Halogenation





Formation of Haloalkane by Addition of Halogen to an Alkene

$$CH_3CH_2Br + NaI \xrightarrow[\text{Heat}]{\text{Acetone}} CH_3CH_2I + NaBr$$

Iodoethane

FINKELSTEIN REACTION

$$2CH_3CH_2Cl + Hg_2F_2 \longrightarrow 2CH_3CH_2F + Hg_2Cl_2$$

Mercurous fluoride

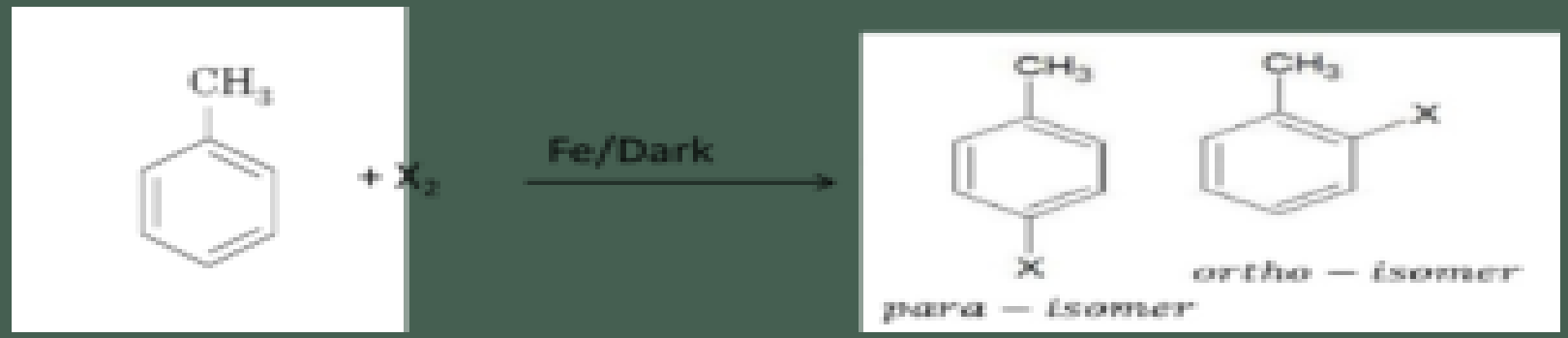
Fluoroethane

SWARTS REACTION

Preparative methods of haloarenes

Direct Halogenation of aromatic Ring

Halogenation of arenes is an electrophilic substitution reaction. The function of the halogen carrier is to generate 'electrophile' which attacks the benzene ring to form the product



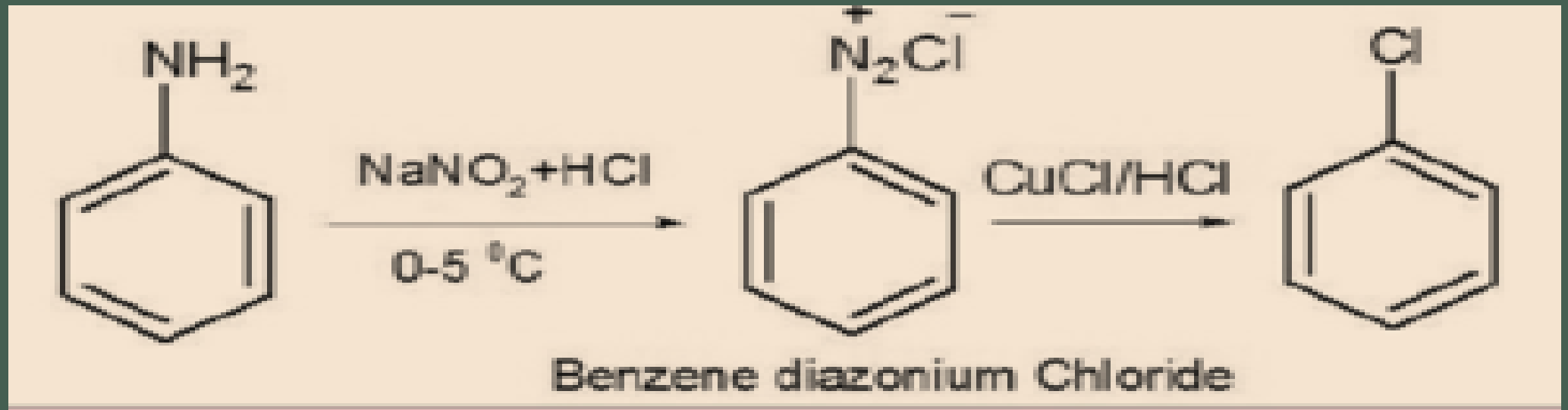
Preparation of aryl chloride and bromide is possible from above reaction.
Aryl fluoride formation is not possible due to the high reactivity of the halogen fluorine.
Additionally, iodine reaction is also not possible as iodine is reversible in nature. Thus, it requires a strong oxidizing agent such as conc. HNO_3 or HIO_4 for the oxidation of HI and converting it to I_2. Therefore, driving the reaction in forwarding direction is difficult and requires a strong oxidizing agent

$$5HI + HIO_3 \longrightarrow 3H_2O + 3I_2$$

Iodic acid

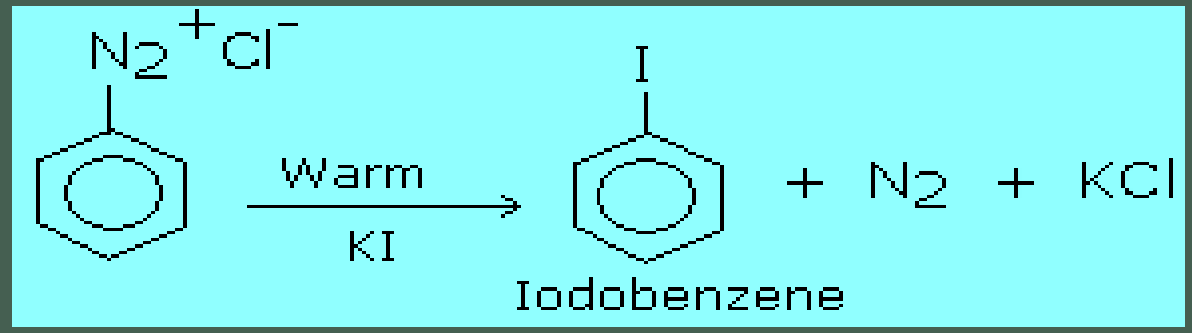
Sandmeyer Reaction

It is a two-step method which includes:
Diazonium salt formation
Diazonium salt reaction with a cuprous halide (Cu_2X_2) (X : Cl, Br)

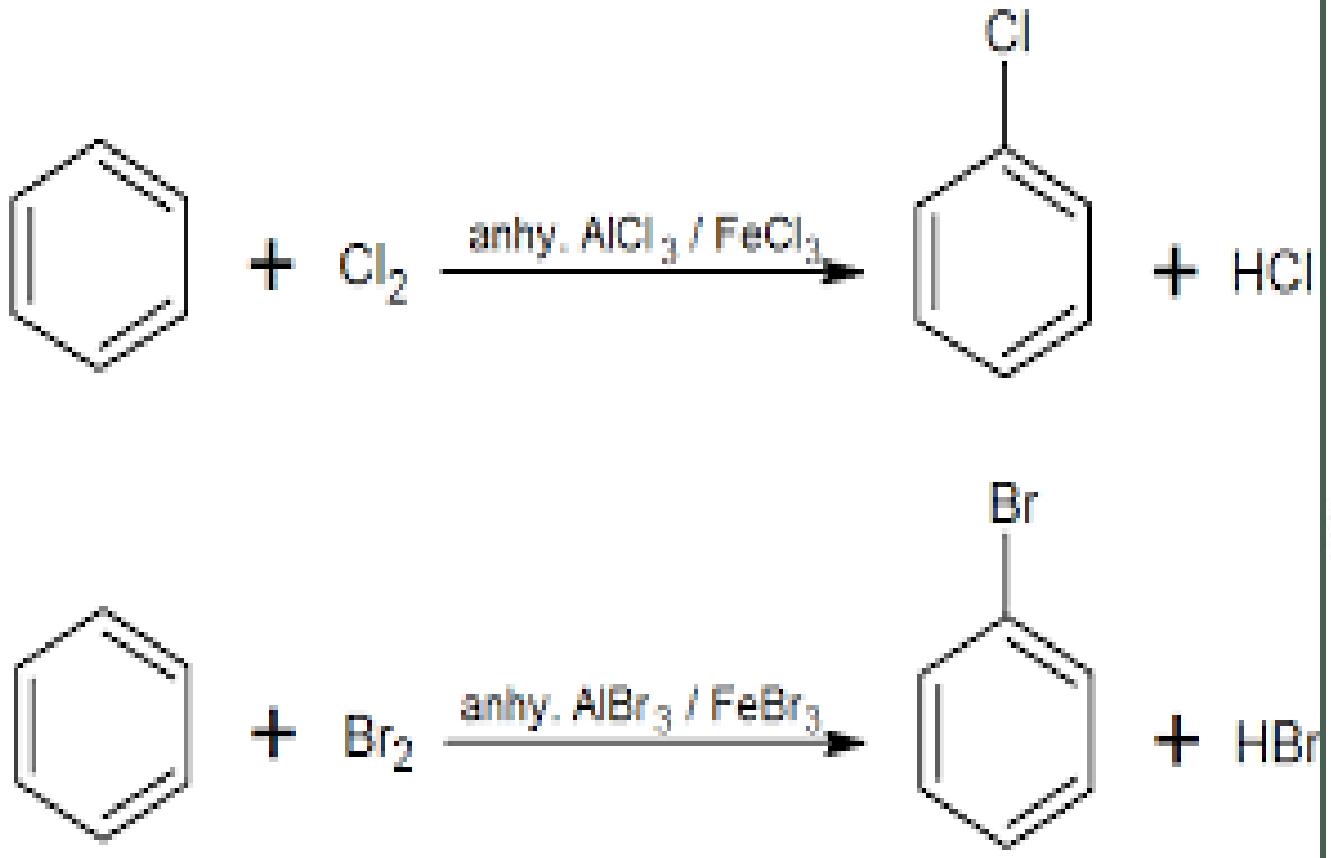


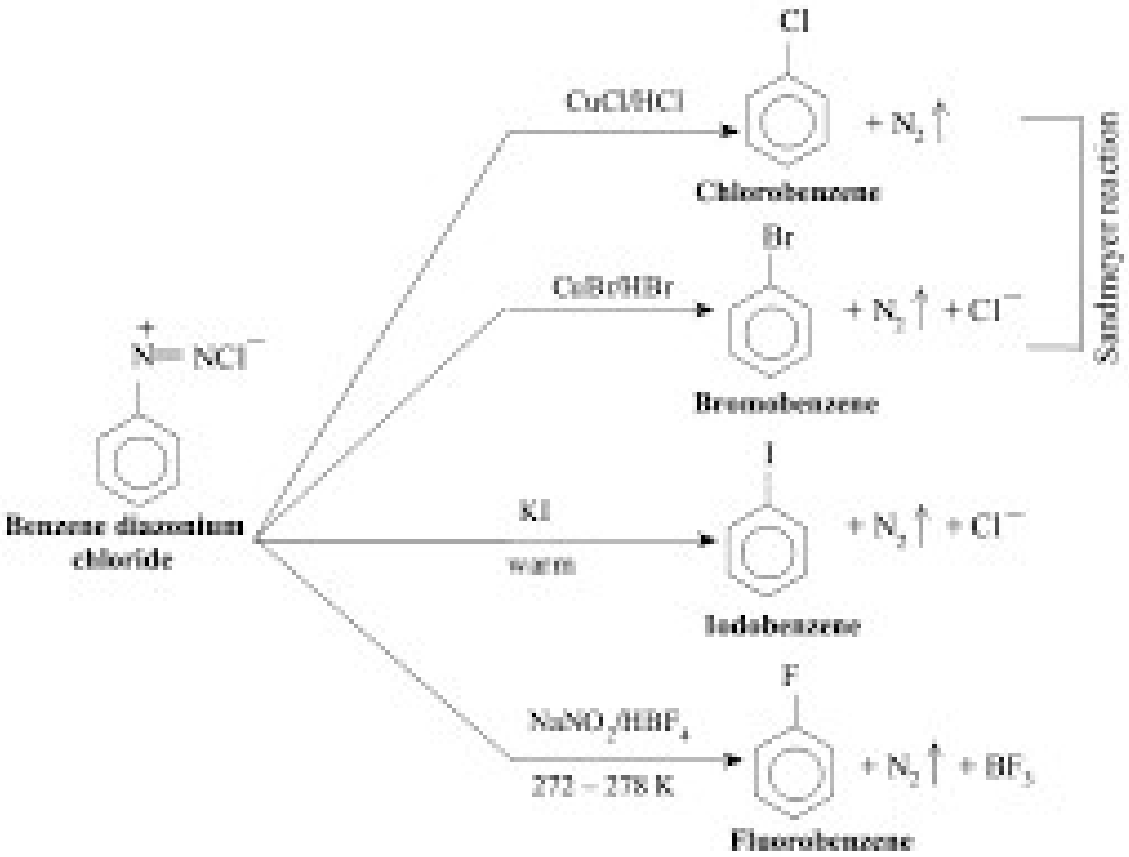
Sandmeyer's Reaction

Replacement of diazonium group by iodine does not require the presence of cuprous halide and is done by simply shaking the diazonium salt with KI.



Preparations of Haloarenes





Physical Properties of Haloalkanes and Haloarenes

1) Physical State

- Alkyl halides are colourless in nature in its pure state.
- However, bromides and iodides develop colour on exposure to light.
- The reason for the development of colour is the decomposition of alkyl halide in presence of light.
- The reaction representing the phenomenon is

$$2R{-}I \rightarrow R{-}R + I_2$$

- Haloarenes are also colourless liquids or crystalline solids that have a characteristic smell.

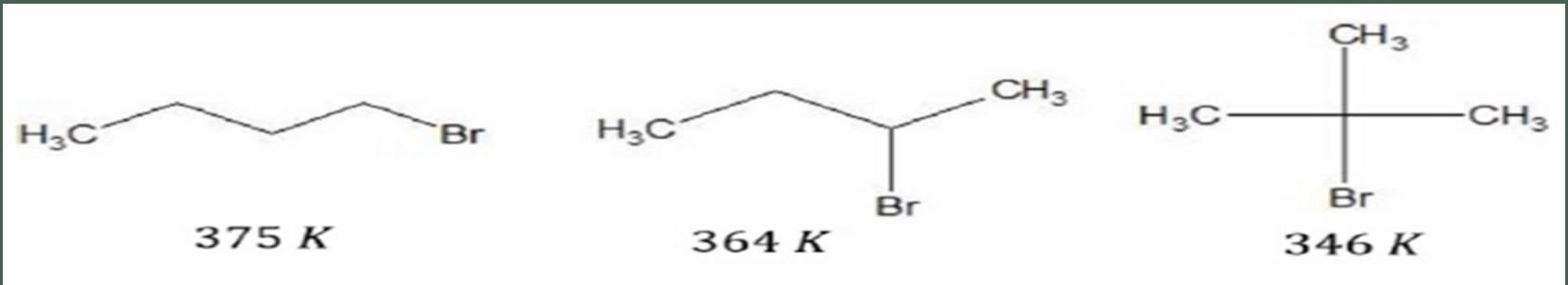
2) Boiling Point

- Boiling points of alkyl halides are higher than those of hydrocarbons having comparable molecular mass.
- The polarity of the C-X bond and higher molecular mass in comparison to the actual hydrocarbon results in the development of very strong intermolecular forces of attraction in the derivatives of halogen.
- The size and molecular mass of halogen members increase when we move down the group in the homologous series thereby forming stronger forces of attraction.
- Hence the boiling point increases as we move down the group in the homologous series.
- **The order of the boiling point of alkyl halides are RI > RBr > RCl > RF**

This is because branching of haloalkanes results in the lesser surface area, thus decreasing the vander Waal's forces of interaction.

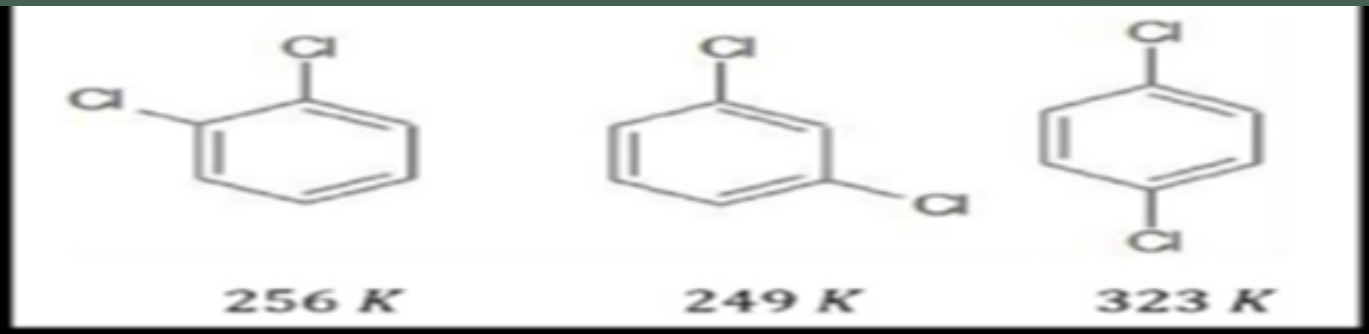
Moreover, as the branching increases the molecule forms sort of a spherical shape resulting in the decrease in the area of contact and forming weaker intermolecular forces.

Boiling points of haloarenes follow the order: **Iodoarene > Bromoarene > Chloroarene**



3) MELTING POINT

- The melting point is based on the strength of the lattice structure of a compound.
- Para-isomer have a higher melting point in comparison to ortho-isomer and meta-isomer of the same compound.
- It is because para-isomers have highly compact crystal lattice in comparison to ortho-isomer and meta-isomer.
- Therefore, higher numbers of molecules are packed compactly in the crystal lattice. Hence, higher energy is required to break the lattice structure thereby increasing the melting point temperature of the compound.



DENISTY

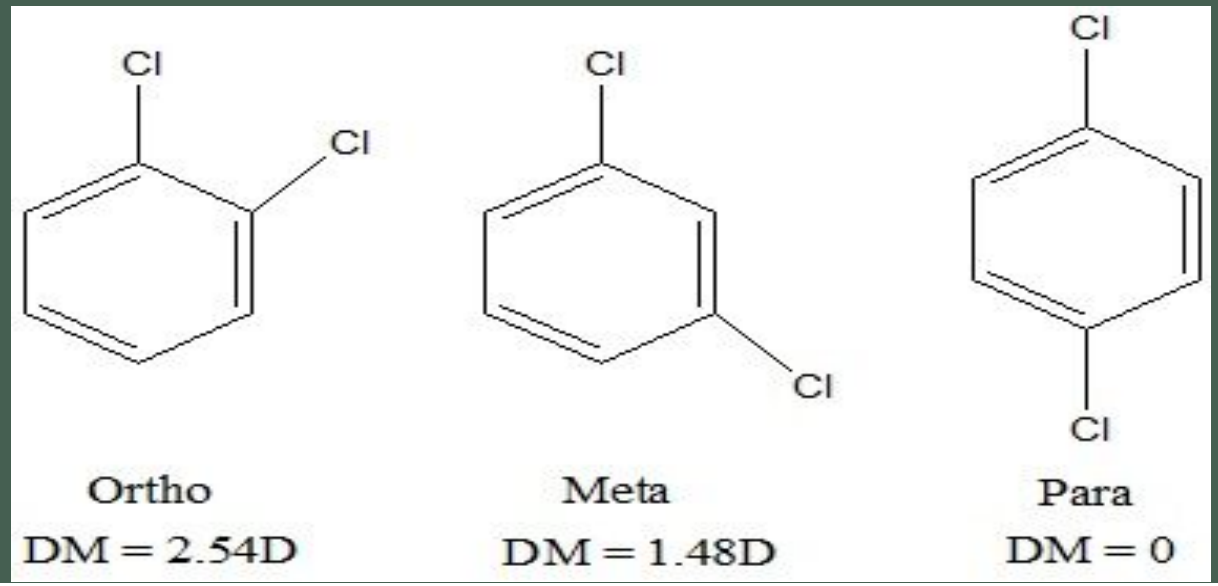
- Density is directly proportional to the mass of any compound.
- Therefore, as the mass increases down the homologous series, the density increases.
- Thus, the derivatives of fluorine are less dense than derivatives of chlorine and derivatives of chlorine are less dense than derivatives of bromine.
- Additionally, density increases with the increase in the number of carbon and halogen atoms.
- Furthermore, it depends upon the atomic mass of the halogen atom. For example, refer to the below diagram

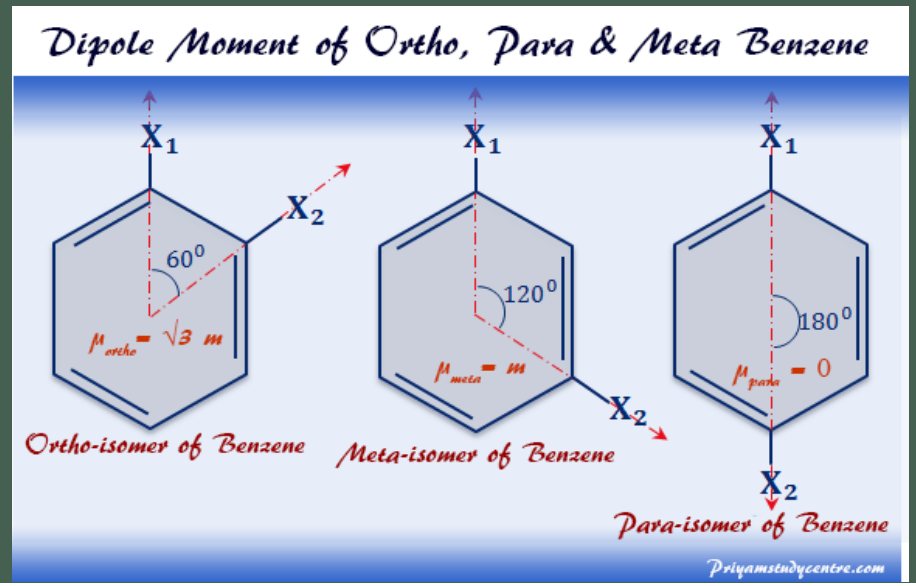
$$n - C_3H_7Cl < n - C_3H_7Br < n - C_3H_7I$$

0.89 g/mL 1.33 g/mL 1.75 g/mL

DIPOLE MOMENT

- The p-isomer with an angle of 180º being more symmetrical has zero dipole moment , the o-isomer with a bond angle of 60º has the highest dipole moment while m-isomer with a bond angle of 120º has an intermediate dipole moment.





SOLUBILITY

- Alkyl halides are slightly soluble in water.
- Even though haloalkanes and haloarenes are polar compounds, they are immiscible with water. Relatively larger amount of energy is required for dissolution of a compound and to break the attractive forces between halogen and the carbon atom.
- However, less energy is released when a bond is formed after dissolution ion and water.
- Moreover, the stability of R-X bond is very low in comparison to the bond formed in water molecules polarity difference
- Thus, haloalkanes and haloarenes neither develop new H-bond nor it breaks the old H-bonds.
- Hence, the solubility of R-X is low. However, these compounds are soluble in organic solvents because of the low polarity of organic solvents such as ether, benzene etc.
- Therefore, the strength of the intermolecular forces between organic solvents and carbon-halogen compounds is similar to carbon-halogen molecules and solvent molecules.

HOME WORK

The best reagent for converting an alcohol into the corresponding chloride is

(a) PCl_5
(b) PCl_3
(c) $HCl/ZnCl_2$
(d) $SOCl_2$

HOME WORK

A hydrocarbon of molecular formula C_6H_{14} on monochlorination gives two products. Identify the structure of hydrocarbon.

THANK YOU STUDENTS

KEEP
LEARNING
STAY SAFE